

Creative Problem Solving [MC6201]**Spring Semester, 2025**

Department	Mathematics and Computer Science	Credits	3	Instructor	Hyunwoo Jung	Class Room	Changjo 702
Subject	Creative Problem Solving	Class Hours/Wk	3	E-mail			
				aryonhwjung@ksa.kaist.ac.kr			
Office Phone	051-606-2350						
Office Hour	2 and 3 period on Monday at Changjo 703						

Instructor would be available at any time after a prior appointment.

1. Course Description

Course Objectives	
This course is a general introduction to creative problem solving. Students learn basic techniques to solve problems by using algorithms and self-directed methods. This course is for self-directed and self-motivated students. For given problems, students try to extract their own creative ideas and write, present, and implement it. Through exploration of this course, students challenge to real world problems. Students define problems formally, extract creative ideas, verify the idea, then implement the idea to apply to real situations.	
Contents	
1. Course Introduction i. Introducing Problem Solving	6. KSA Timetable Construction Algorithm i. Explaining the Ideas ii. How to Implement
2. CPS for Problem Set #1 i. Presentations of Ideas ii. Discussion for the Ideas	7. Defining Problems i. Formalize Real-World Problems ii. Well-defining the Problem iii. Present defined problems
3. Problem Presentations i. Individual Presentation ii. Modifying the Problems	8. Deriving Ideas i. Design Efficient Algorithms
4. Graph Algorithm Techniques i. DFS, BFS ii. Bi-connected Component	9. Present Final Results
5. CPS for Problem Set #2 i. Presentations of Ideas ii. Discussion for the Ideas	10. Supplement the Ideas and Implementation
	11. Final Modifications

2. Core Abilities

Course Core Abilities	
- Problem Solving, Inference, Communication, Attitude and Tenacity, Problem Finding and Definition, Ethics	
- Problem Finding and Definition	Students try to find their own real life problems by themselves and try to define formally
- Inference	Students develop themselves to understand abstract concepts and generalize them
- Problem Solving	Students devise their creative diverse ideas to tackle problems
- Communication	Students communicate with other peer students and the teacher through active presentations
- Attitude and Tenacity	Students learn basic attitude to become scientist and importance of tenacity through problem solving process
- Ethics	Students try to tackle the problems by themselves without dishonest actions. Students learn taking responsibility of dishonest actions.

3. Prerequisites

Basic programming skill that is acquired through information science 1 and 2 courses is required. Logical Thinking. Analytical Thinking. 99% Perspiration.

4. Course Info

There is no main textbook

Main materials can be downloaded from Google classroom course link:

5. Grading Policy

Homework Assignments	Midterm Exam	Individual Project	Attendance & Attitude
20%	40%	30%	10%

1. Homework Assignment

A. Each student presents solutions for each problem set

2. Individual Project

A. Each student define real-world problem, derive algorithm, implement it, finally apply to real world situation

3. Attendance & Attitude

A. Bad attitudes such as sleeping, chatting, etc. result in deduction

6. Weekly Schedule

Weeks	Topics
Week1	Course Introduction and CPS for Problems

Week2		CPS Problems
Week3		DFS and BFS
Week4		Graph Algorithm Techniques
Week5		Graph Algorithm Techniques
Week6		CPS for Problems
Week7		Wrap-up
Week8		Midterm Exam
Week9		KSA Timetable Construction Algorithm
Week10		Defining Problems
Week11		Deriving Ideas
Week12		Present Final Results
Week13		Supplement the Ideas and Implementation
Week14		Final Modification
Week15		Supplement the Ideas and Implementation
Week16		Final Exam Period

*** The schedule can be changed as it may chance.

7. Homework submission due date

There is a deadline for each assignment.

8. Policy for cheating

No cheating activity is allowed. Students who cheat, plagiarize, copy, fake, etc. will fail the course.

9. Proportion of A or B Grades

Basically, the proportion of students who get A or B grade is 70%. But if all students get high achievements, this ratio would be raised.

Aim of this course is for you to make you exploring problem solving. If you have any difficulty for taking this course, try to get advices from the instructor.